BOARD OF INTERMEDIATE AND SECONDARY EDUCATION

Annual Examination 2026

Chemistry — SSC-I (9th Grade)

Syllabus: Chapter 9 — Chemical Equilibrium

Total Marks: 65	**Time Allowed: 2h 30m**	**Version: 1**

General Instructions:
1. Write your name, roll number and section clearly on the answer sheet.
2. Section A is compulsory. Encircle the correct option on the question paper itself.
3. In Section B, attempt **all 11** questions. For each question, attempt either the main question **OR** its alternative.
4. In Section C, attempt **all 4** questions. For each question, attempt either the main question **OR** its alternative.
5. Use black or blue ink only. Pencil is not allowed except for diagrams.

SECTION A — Multiple Choice Questions (12 × 1 = 12 Marks)

#	Question	A	B	C	D	Answer
1	Which is true about the equilibrium state?	Forward reaction stops	Reverse reaction stops	Both reactions stop	Both continue at same rate	ⒶⒷⒸⒹ
2	A reversible reaction is represented by the symbol:	$\rightarrow$	$\leftrightarrow$	$\rightleftarrows$	$\rightleftharpoons$	ⒶⒷⒸⒹ
3	In an irreversible reaction, equilibrium is:	established quickly	established slowly	never established	established when reaction stops	ⒶⒷⒸⒹ
4	Which does NOT happen when a system reaches equilibrium?	Both reactions stop	Rates become equal	Concentrations stop changing	Reaction continues in both directions	ⒶⒷⒸⒹ
5	Hydrated copper(II) sulphate $CuSO_4{\cdot}5H_2O$ is:	a white solid	a blue solid	a pink solid	a colourless liquid	ⒶⒷⒸⒹ
6	When $CuSO_4{\cdot}5H_2O$ is heated, the product formed is:	$CuSO_4$ (white solid)	CuO (black solid)	Cu (red metal)	$CuSO_3$ (grey solid)	ⒶⒷⒸⒹ
7	Anhydrous cobalt(II) chloride $CoCl_2$ is:	a pink solid	a blue solid	a white solid	a green solid	ⒶⒷⒸⒹ
8	When water is added to anhydrous $CoCl_2$, the colour change observed is:	white to blue	blue to pink	pink to white	blue to white	ⒶⒷⒸⒹ
9	$H_2 + I_2 \rightleftarrows 2HI$ is sealed at 25°C and equilibrium is established. Substances present are:	H_2 and I_2 only	HI only	H_2 only	H_2, I_2 and HI	ⒶⒷⒸⒹ
10	Concentrations at equilibrium remain unchanged provided:	concentration alone is unchanged	temperature alone is unchanged	pressure alone is unchanged	all three conditions are maintained	ⒶⒷⒸⒹ
11	Heating $CoCl_2{\cdot}6H_2O$ shifts the equilibrium to the right because:	heating always favours the reverse reaction	the forward (dehydration) reaction is endothermic	heating decreases rates of both reactions equally	water vapour concentration increases, shifting equilibrium left	ⒶⒷⒸⒹ
12	Chemical equilibrium is called *dynamic* because:	concentrations are always changing	only the forward reaction occurs	reactions continue but concentrations remain constant	the system is in a closed container	ⒶⒷⒸⒹ

SECTION B — Short Questions (11 × 3 = 33 Marks)

Attempt all 11 questions. For each pair, attempt either the main question OR its alternative.

Part	Question	Marks	OR	OR Question	Marks
2(i)	Define a *reversible reaction*. Give two examples from the textbook with the symbol used.	**3**	**OR**	Define a *complete reaction*. How does it differ from a reversible reaction? Give one example of each.	**3**
2(ii)	Define *chemical equilibrium*. Why is it called *dynamic*? State two characteristics of the equilibrium state.	**3**	**OR**	Using $2SO_{2(g)} + O_{2(g)} \rightleftarrows 2SO_{3(g)}$, explain how concentrations change from the start until equilibrium is reached.	**3**
2(iii)	State the three conditions necessary for a chemical system to maintain equilibrium (Section 9.2).	**3**	**OR**	What is a *closed system*? Why must equilibrium reactions occur in a closed system? Give one example.	**3**
2(iv)	Describe the experiment with $CuSO_4{\cdot}5H_2O$. State the colour changes observed. Write the reversible equation.	**3**	**OR**	How can anhydrous $CuSO_4$ be used as a test for water? What colour change confirms water? Write the equation.	**3**
	Describe the equilibrium of $CoCl_2{\cdot}6H_2O$. State			Write the reversible equation for the	

2(v)	the colours of hydrated and anhydrous forms. What happens when the pink solid is heated?	**3**	**OR**	$CoCl_2 \cdot 6H_2O$ equilibrium. What happens to colour and equilibrium when water is added to anhydrous $CoCl_2$?	**3**
2(vi)	How does increasing temperature affect a chemical equilibrium? In which direction does equilibrium shift and why? Use $CuSO_4 \cdot 5H_2O$ as an example.	**3**	**OR**	Predict the direction of shift when $CoCl_2 \cdot 6H_2O$ equilibrium is cooled. Explain using the principle that the system opposes the change.	**3**
2(vii)	For $CO_{(g)} + 3H_{2(g)} \rightleftarrows CH_{4(g)} + H_2O_{(g)}$, write the forward reaction and the reverse reaction separately.	**3**	**OR**	Bromine chloride (BrCl) decomposes to form Br_2 and Cl_2. Write the reversible equation. Identify the forward reaction and state what conditions favour the reverse reaction.	**3**
2(viii)	For $N_{2(g)} + 3H_{2(g)} \rightleftarrows 2NH_{3(g)}$, identify (a) the forward reaction and (b) the reverse reaction. State the catalyst and conditions from the textbook.	**3**	**OR**	For $2SO_{2(g)} + O_{2(g)} \rightleftarrows 2SO_{3(g)}$, identify (a) forward and (b) reverse reactions. State the catalyst and industrial conditions.	**3**
2(ix)	Using the fizzy drinks example (Science Titbits), explain how CO_2 demonstrates physical equilibrium. Write the equation and state what happens when the lid is removed.	**3**	**OR**	Explain how placing a liquid in a closed container leads to equilibrium between liquid and vapour. Write the equilibrium equation.	**3**
2(x)	Differentiate between forward and reverse reactions. Use $N_{2(g)} + 2O_{2(g)} \rightleftarrows 2NO_{2(g)}$ as an example, writing each separately.	**3**	**OR**	In $2NO_{2(g)} \rightleftarrows N_{2(g)} + 2O_{2(g)}$, identify which reaction produces products and which restores reactants. Compare the rates of these two reactions at equilibrium.	**3**
2(xi)	Name two important chemicals mentioned in the Chapter 9 introduction whose industrial production involves equilibrium reactions. Why is understanding equilibrium important in industry?	**3**	**OR**	State any four Key Points about chemical equilibrium from Chapter 9 Key Points section (p. 138).	**3**

SECTION C — Long Questions (4 × 5 = 20 Marks) — Attempt all 4

Part	Question	Marks	OR	OR Question	Marks
3(i)	(a) Define reversible reaction and chemical equilibrium. Distinguish between a complete and reversible reaction with one example each. [3] (b) State the three conditions for equilibrium. What happens to equilibrium if one condition changes? [2]	**5**	**OR**	(a) Using $2SO_{2(g)} + O_{2(g)} \rightleftarrows 2SO_{3(g)}$, explain step by step how equilibrium is established: initial state, how rates change, and final equilibrium state. [3] (b) Why is equilibrium called *dynamic*? Distinguish dynamic from static equilibrium. Give a real-life analogy. [2]	**5**
3(ii)	(a) Describe the laboratory experiment on $CuSO_4 \cdot 5H_2O$: materials, procedure (3 steps), observations, and conclusion. [3] (b) Write the reversible equation. Label the forward and reverse directions and state the colour of each compound. [2]	**5**	**OR**	(a) Describe the $CoCl_2$ equilibrium: state colours of hydrated and anhydrous forms, and explain what happens when (i) heated and (ii) water is added. [3] (b) Write the reversible equation. Which direction does equilibrium shift on heating? Explain using the endothermic principle. [2]	**5**
3(iii)	(a) State the general principle for the effect of temperature on equilibrium. Apply it to $CoCl_2 \cdot 6H_2O \rightleftarrows CoCl_2 + 6H_2O$: predict the shift when (i) heated and (ii) cooled. [3] (b) Explain the effect of adding water to the $CoCl_2$ equilibrium: direction of shift and reason. [2]	**5**	**OR**	(a) For $N_{2(g)} + 3H_{2(g)} \rightleftarrows 2NH_{3(g)}$, write the forward and reverse reactions separately. State the catalyst, temperature, and pressure from the textbook. [3] (b) Define closed system. Why does an open system prevent equilibrium? Give one example. [2]	**5**
3(iv)	(a) State all the Key Points of Chapter 9 as listed in the textbook (at least six required for full marks). [3] (b) Compare reversible and irreversible (complete) reactions under: (i) direction, (ii) equilibrium, (iii) textbook example, (iv) symbol. [2]	**5**	**OR**	(a) Explain the liquid-vapour equilibrium (Liquid ⇄ Vapours): describe the process from start to equilibrium and explain what is meant by evaporation rate equalling condensation rate. [3] (b) Using the CO_2 fizzy drinks example, explain how pressure affects equilibrium. State the equation and what happens when (i) lid is sealed and (ii) lid is removed. [2]	**5**

Invigilator's Signature: ____________________ Candidate's Signature: ____________________

Chapters Covered: Chapter 9 — Chemical Equilibrium (pp. 135–139)

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